PROVISIONAL PATENT APPLICATION

Controlled Relational-Interior Interface, Separately Maintained Interior Address Network, Custody-Preserving Threshold, and Methods of Qualifying, Operating, and Safely Releasing the Same

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This specification is complete and self-contained. Applicant's companion applications are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no non-Euclidean occupied interior, exterior-to-interior custody transfer, anomalous capacity, or qualified interface is represented as achieved.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to controlled angular-momentum coherence, boundary control, completion-address engineering, and applications thereof. Companion applications concern support-force control, transportation, media and arrays, information, rate control, completion-scheduled computation, molecular control, boundary translation, direct-adjacency transport, and counterfactual receipt preservation. Each application is self-contained. This application is directed specifically to a threshold that joins an exterior boundary address book to a separately maintained interior relational network while preserving custody, interface capacity, and safe release.

FIELD

Controlled interfaces between separately maintained relational networks; phase-locked network hardware; reconfigurable adjacency and path-cost systems; custody-preserving admission and release; and occupied interior systems whose operational address capacity, path metric, or topology is not required to embed isometrically within the exterior metric display of the interface housing.

BACKGROUND

1. Conventional enclosures are treated as subsets of a surrounding Euclidean volume. Their usable volume, path geometry, and capacity are consequently bounded by the displayed exterior hull, except for ordinary folding, compression, remote storage, optical illusion, or virtual representation.

2. Network systems, waveguide networks, switched fabrics, phase arrays, resonator lattices, and distributed clocks already demonstrate that operational adjacency and path cost need not match physical drawing distance. Existing systems, however, ordinarily treat that mismatch as signal routing inside a larger container and do not provide an occupied threshold with a declared custody invariant.
3. The applicant identifies a distinct engineering architecture: an exterior interface network and an interior network are maintained as separate address books; an interface state selects a temporary cross-book adjacency; a custody controller admits an object or retained state only after both books co-sign the transfer; and entry, interior persistence, exit, support energy, and every transformed invariant are recorded.
4. The architecture has an immediately buildable rung. Phase-locked oscillators, resonators, switched links, delay lines, sensor-tagged tokens, robotic carriers, and reconfigurable graphs can implement and test the interface protocol, topology control, route-order dependence, holonomy, throughput, failure morphology, and safe release. The stronger embodiment—in which an occupied interior has an operational metric or capacity not bounded by the exterior hull's displayed metric volume—requires a physical address-transfer law not presently established.
5. What the art lacks, and this application supplies, is a self-contained system tying: (i) separately maintained exterior and interior adjacency records; (ii) an interface-state controller; (iii) a custody token carried continuously through admission, interior operation, and release; (iv) phase and topology witnesses; (v) a measured interface-response map; (vi) finite-capacity and support ledgers; and (vii) null-valid qualification gates that distinguish a genuine cross-book operation from hidden ordinary routing, remote substitution, display manipulation, destructive reconstruction, or software-only simulation.

SUMMARY

The disclosure is organized in three tiers.

Tier 1—product. A controlled relational-interior system (100; FIG. 1) includes an exterior boundary assembly, an interior network assembly, an interface coupler, phase references, address and custody registries, an interface-state controller, witness sensors, a support ledger, and a safe-release subsystem. The controller admits a transfer only when a stored qualified interface map authorizes the requested exterior address, interior admission address, interface state, payload class, and operating envelope.

Tier 2—qualification. A qualification fixture (600; FIG. 6) measures adjacency, delay, phase, route order, custody continuity, capacity, support cost, leakage, and hidden-channel witnesses under active, sham, bypass, relabeling, duplicate-token, phase-reversal, topology-reversal, and interruption states. The output is a qualified map or a scoped null certificate.

Tier 3—occupied prophetic branch. A physical threshold maintains an interior adjacency book independently of the exterior display and carries a retained organization across the interface without destruction or duplicate instantiation. This branch is claimed as a prophetic apparatus and method; no successful occupied non-embeddable interior is represented.

The product is useful at the realizable rung as a reconfigurable phase-network testbed, secure custody gate, topology-switching fabric, distributed robotics interface, and metrology platform for relational path control. The same architecture preserves priority for later occupied embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Controlled relational-interior system with exterior assembly, interior network, interface coupler, custody registry, controller, witnesses, support ledger, and safe release.
- FIG. 2 — Exterior and interior address books joined by a commanded interface edge while remaining separately maintained.
- FIG. 3 — Admission, interior-operation, and release state machine with custody continuity and fault quarantine.
- FIG. 4 — Weight-control and topology-control modes, route-order comparison, and holonomy receipt.
- FIG. 5 — Finite interface-capacity, occupancy, queue, support-cost, and graceful-degradation account.
- FIG. 6 — Qualification fixture and randomized countermodel matrix.
- FIG. 7 — Empirical interface map and layered scope from raw network receipt to occupied prophetic operation.
- FIG. 8 — Reference phase-network demonstrator using resonator or oscillator nodes, switched links, delay/phase metrology, and tagged carriers.
- FIG. 9 — Prophetic occupied threshold, interior custody set, and independent interior support architecture.
- FIG. 10 — Qualification-to-deployment flow, provenance binding, validity states, and recalibration loop.

DEFINITIONS

1. **Address** — an operationally distinguishable participation state in a declared network, fixed by a registry and observable completion conditions rather than by three coordinates alone.
2. **Address book** — the authenticated record of nodes, admissible links, link costs, phase references, capacities, and validity states for one network.
3. **Exterior network** — the network containing the externally observed boundary addresses of the housing or threshold.

4. **Interior network** — a separately maintained network containing interior nodes and links. Separate maintenance requires an independently identified registry, phase reference or phase-transfer rule, support account, and validity state; a visual image stored in the exterior controller alone is not an interior network.
5. **Interface edge** — a controller-enabled relation joining a selected exterior address to a selected interior address under declared admission and release rules.
6. **Interface state vector** — measured values sufficient to specify an attempted interface operation, including selected addresses, link state, phase offset, coupling amplitude, route order, timing, occupancy, payload class, environmental state, and support input.
7. **Custody token** — a unique, nonduplicated physical or cryptographically bound record associated with the admitted object or retained state and updated at every transfer boundary. In an occupied embodiment the token is supplemented by continuous physical witnesses of the declared retained invariants.
8. **Custody continuity** — successful reconciliation of the unique token and declared retained invariants across admission, interior operation, and release, without a simultaneous duplicate, unexplained disappearance, or destructive reconstruction substituted for transfer.
9. **Operational metric** — the shortest accumulated qualified link cost between addresses under a frozen address book and cost rule. Cost may comprise time, phase, energy, loss, required completions, or a calibrated combination.
10. **Weight control** — adjustment of one or more existing link costs while retaining the link set.
11. **Topology control** — addition, removal, or rerouting of admitted links.
12. **Holonomy receipt** — the measured difference in retained phase, orientation, timing, or another declared invariant after two ordered interface routes having the same nominal endpoints.
13. **Interface capacity** — the measured payload, transaction, address, energy, bandwidth, occupancy, or custody rate supportable while all validity gates remain satisfied.
14. **Support ledger** — synchronized measurement of energy, momentum, angular momentum, timing, heat, control traffic, storage, and other support accounts required by the declared operation.
15. **Empirical interface map** — a stored relation between interface state vector and measured outputs, including uncertainty, provenance, permissible interpolation region, capacities, failure modes, and safe-release parameters. A map may contain only ordinary-routing or null results.
16. **Qualified relational interface** — an interface having a map that passes the applicable gates K1–K10 for its declared scope. Qualification at a network-demonstrator scope does not establish occupied non-embeddable geometry.
17. **Non-embeddable interior claim** — a claim that the measured interior operational metric, topology, or independently occupied address capacity cannot be represented as an ordinary subset, folded path, remote ordinary facility, display, or software simulation inside the declared exterior reference model.

18. **Safe release** — a controller state that stops new admissions, preserves custody records, returns admitted carriers or states through a qualified exit when possible, isolates unresolved custody, removes commanded interface edges in a stored order, and preserves the final support ledger.
19. **Validity states** — qualifying, valid, degraded, out-of-envelope, invalid, quarantine, and safe release.
20. **Countermodel** — an ordinary mechanism capable of imitating an asserted interface receipt, including hidden wired or wireless routing, undisclosed remote storage, duplicate substitution, delayed release, display manipulation, relabeling, destructive reconstruction, or common-clock artifact.

THEORETICAL FOUNDATION AND OPEN TRANSFER QUESTION

The motivating POAMS framework treats projected space as an address display assembled from completed relations rather than as a conserved container. A retained organization may therefore have an exterior address relation distinct from its internal constitution. Graph mathematics establishes only the formal possibility: for a connected weighted graph, adding or repricing an edge changes the operational metric without changing node identity. It does not establish that a material payload can be transferred between independently maintained physical address books.

Let the exterior and interior networks be

$G_E=(V_E,E_E,w_E)$ and **$G_I=(V_I,E_I,w_I)$**,

with a commanded interface edge **$e=(a_E,a_I,w_e,\chi)$** controlled by interface state **χ**. The combined operational metric is the shortest qualified accumulated cost in **$G_E \cup G_I \cup \{e\}$**. This is a descriptive construction. The open physical transfer question is whether a boundary apparatus can make **e** a lawful completion available to a retained material organization while preserving custody.

The qualification program therefore measures an open relation

$Y = K(\chi,$ payload, phase, route, occupancy, environment, support),

where **Y** contains adjacency, delay, phase, custody, capacity, leakage, and support receipts and where K may describe only conventional routing. No scale law or non-embeddable result is presumed.

PRODUCT ARCHITECTURE (100; FIG. 1)

The exterior boundary assembly (110) contains at least one exterior address port and witnesses its local timing, phase, energy, and custody state. The interior assembly (120) contains a plurality of nodes joined by qualified links, an interior registry, and independent support sensing. The interface coupler (130) is a switchable optical, acoustic, radio-frequency, mechanical, photonic, magnonic, superconducting, electronic, or other phase-bearing coupling structure. A reference demonstrator uses phase-locked oscillator or resonator nodes connected by calibrated switched links.

The registry and custody store (140) binds node identities, allowed routes, cost definitions, payload classes, unique custody tokens, and provenance. The interface controller (150) selects an interface edge, commands phase and coupling, checks capacity, and admits transfer only inside the stored interface map. Witnesses (160) measure both sides of the threshold and every known bypass. The support ledger (170) records the complete run. Safe release (180) removes the interface in a stored sequence and returns or quarantines custody.

DEPLOYED OPERATING METHOD (FIG. 3)

The controller: (a) authenticates the exterior and interior address books; (b) verifies map provenance and validity; (c) assigns a unique custody token; (d) measures baseline routes and bypass witnesses; (e) selects the interface state; (f) obtains co-signing readiness from both endpoint assemblies; (g) enables the interface edge; (h) admits the declared carrier or state; (i) confirms terminal custody and removes source custody according to the declared transfer class; (j) monitors interior operation and capacity; (k) performs release through the same or another qualified interface; and (l) reconciles token, invariants, and support ledger. A failed reconciliation enters quarantine or safe release.

LAYERED SCOPE (FIG. 7)

- **L0:** raw phase, delay, route, custody, energy, and environmental records.
- **L1:** qualified network adjacency or path-cost control in a conventional demonstrator.
- **L2:** cross-book interface operation with unique-token custody and countermodel bounds.
- **L3:** transfer of a retained physical organization between separately maintained networks.
- **L4:** occupied interior whose measured operational metric or capacity is not bounded by the exterior displayed metric volume.

No lower layer establishes a higher layer.

QUALIFICATION AND ACCEPTANCE GATES (FIG. 6)

The fixture randomizes active interface, sham coupling, ordinary bypass, relabeling, duplicate-token, phase reversal, topology reversal, route-order reversal, delayed-release, and interruption states. Qualification requires:

- **K1 identity:** authenticated node and payload identities remain bound to provenance;
- **K2 adjacency contrast:** the selected edge changes the declared operational metric relative to matched sham states;
- **K3 phase receipt:** measured phase and timing agree with the selected edge model within uncertainty;
- **K4 custody uniqueness:** one token and one physical custody record exist throughout;
- **K5 bypass exclusion:** instrumented ordinary paths cannot account for the receipt;

- **K6 order/holonomy:** predicted route-order receipts repeat or a scoped null is recorded;
- **K7 capacity:** occupancy and throughput remain within measured limits;
- **K8 support closure:** energy, momentum, angular momentum, heat, timing, and control traffic reconcile;
- **K9 interruption safety:** power loss, phase slip, link loss, and controller fault enter a bounded state with recoverable custody; and
- **K10 independent confirmation:** a second apparatus, implementation, or laboratory reproduces the scoped result.

A conventional network demonstrator may pass K1–K10 at L1 or L2 while providing no evidence for L3 or L4.

DETAILED EMBODIMENTS

E1—phase-network demonstrator (FIG. 8). Sixteen to 256 phase-addressable nodes are implemented as coupled resonators, oscillators, photonic delay elements, superconducting resonators, software-defined radios with physical timing ports, or combinations thereof. Each link has a measured delay, phase, loss, capacity, and support cost. A movable instrumented token or robotic carrier supplies a physical custody channel. The interface controller creates and removes cross-book links under randomized trials. This embodiment is buildable with extant components and validates control logic, metrology, failure recovery, and countermodel handling.

E2—independent-support interior. The interior assembly has its own power storage, clock or phase-transfer unit, environmental control, structural support, and witness array. The interface transmits only declared custody and support exchanges. Removing exterior controller power does not silently erase the interior record; the resulting behavior is measured and classified.

E3—topology and holonomy instrument. The controller executes A→B→C and A→C→B routes using identical endpoint labels and compares retained phase, orientation, and timing. Reversal and relabeling controls distinguish ordered physical operation from database labeling.

E4—finite-capacity gate. Admission is limited by measured coupling bandwidth, phase stability, stored energy, heat rejection, occupancy, and custody reconciliation time. The controller queues or refuses admissions before the weakest limit is reached and records graceful degradation rather than allowing uncontrolled recruitment of nodes.

E5—prophetic occupied interior (FIG. 9). A threshold housing contains an exterior boundary array and interface controller. A separately supported interior custody set maintains occupants, equipment, local clocks, internal paths, and environmental services. Entry and exit preserve declared biological, mechanical, clock, memory, and orientation invariants. The embodiment is enabled as an architecture and experimental sequence but depends on a presently open physical interface law; no working device is represented.

PROPHETIC EXAMPLES

P1—32-node bench network. Two eight-node exterior clusters and a sixteen-node interior cluster use measured phase-locked links. The interface controller inserts one cross-book link, transports a uniquely tagged carrier, removes the link, and returns the carrier. Randomized bypass and relabeling trials test K1–K10. Success establishes L1/L2 only.

P2—capacity surface. The controller varies concurrent tokens, link amplitude, phase error, and support power and stores a capacity surface with uncertainty. The operating envelope ends before custody error or phase-slip probability exceeds a frozen bound.

P3—occupied-threshold progression. If an independently reproduced physical interface anomaly survives all ordinary-channel controls, testing progresses from a passive witness, to an encoded retained state, to a nonliving instrument package, and only later to a living payload under separate safety review. No stage is skipped because a lower-layer network demonstration succeeds.

POAMS-NATIVE DIRECT-OPERATION EMBODIMENT

In the prophetic native embodiment the exterior address of a retained organization is a boundary relation rather than a coordinate occupied by all of its internal structure. The interface controller maintains the organization's internal closure complex while changing which exterior and interior address books may co-sign its next completion. The achieved operation is measured; no nonzero transfer authority, anomalous capacity, or safe occupied result is presumed.

CLAIMS

1. A controlled relational-interior system comprising: an exterior boundary assembly having an exterior address registry; an interior network assembly having a separately maintained interior address registry and a plurality of interior nodes joined by qualified links; a controllable interface coupler between a selected exterior address and a selected interior address; a custody registry configured to bind a unique custody token to an admitted carrier or retained state; witness sensors measuring phase, timing, link state, occupancy, support input, and bypass channels on both sides of the interface coupler; a memory storing an empirical interface map with provenance, uncertainty, a permissible operating region, interface capacity, and safe-release parameters; and an interface controller configured to authenticate both registries, select an interface state from the map, enable a cross-registry interface edge, admit the carrier or retained state only after both selected addresses report readiness, reconcile the unique custody token through admission and release, and enter safe release when a validity condition fails.
2. The system of claim 1, wherein the interior network has a phase reference, support supply, and validity state independently identified from those of the exterior boundary assembly.
3. The system of claim 1, wherein the qualified links comprise optical, radio-frequency, acoustic, mechanical, photonic, magnonic, superconducting, electronic, or hybrid phase-bearing links having

measured delay, phase, loss, and capacity.

4. The system of claim 1, wherein the controller performs weight control by changing a measured cost of an existing link without changing the admitted link set.
5. The system of claim 1, wherein the controller performs topology control by adding, removing, or rerouting an admitted link.
6. The system of claim 1, wherein the controller executes two ordered routes having common endpoints and stores a holonomy receipt comprising a difference in phase, orientation, timing, or another retained invariant.
7. The system of claim 1, wherein the interface capacity is bounded by at least two of coupling bandwidth, phase stability, occupancy, custody-reconciliation time, support energy, heat rejection, and leakage.
8. The system of claim 1, wherein safe release stops new admissions, preserves the custody registry, returns admitted custody through a qualified exit when available, and quarantines an unresolved custody record.
9. The system of claim 1, wherein the controller refuses an empirical interface map whose provenance does not match an identifier of the exterior boundary assembly or interior network assembly.
10. The system of claim 1, wherein the carrier is a tagged physical token or robotic carrier and the system is a phase-network qualification demonstrator.
11. A method of operating a controlled interface between separately maintained exterior and interior networks, comprising: authenticating exterior and interior address books; assigning a unique custody token to a carrier or retained state; measuring baseline routes and bypass channels; selecting an exterior address, an interior address, and an interface state from a stored qualified map; obtaining readiness receipts from endpoint assemblies associated with both addresses; enabling an interface edge; admitting the carrier or retained state; confirming terminal custody and updating source custody according to a declared transfer class; monitoring phase, timing, occupancy, support inputs, and bypass witnesses; releasing the carrier or retained state through a qualified interface; and reconciling the unique custody token and a support ledger, with a failed reconciliation causing quarantine or safe release.
12. The method of claim 11, further comprising comparing an active interface state with randomized sham, ordinary-bypass, relabeling, duplicate-token, phase-reversal, route-reversal, delayed-release, and interruption states.
13. The method of claim 11, further comprising constructing an operational metric from accumulated measured link costs and determining a change in the metric caused by the enabled interface edge.
14. The method of claim 11, further comprising varying occupancy and interface state to establish a capacity surface and refusing admission outside a permissible region of the capacity surface.
15. The method of claim 11, further comprising executing two differently ordered routes having the same endpoints and recording a holonomy receipt.

16. The method of claim 11, wherein a null, conventional-routing result, or failed countermodel gate is stored without enabling a higher-layer occupied transfer mode.
17. A qualification apparatus for a controlled relational interface comprising first and second phase-addressable node networks having separate registries; a controllable coupling structure selectively joining one node of the first network to one node of the second network; a uniquely tagged carrier; phase, delay, loss, energy, occupancy, and bypass-channel sensors; a randomized-state controller; and a map writer, wherein the controller executes active, sham, bypass, relabeling, duplicate-token, phase-reversal, topology-reversal, route-order-reversal, delayed-release, and interruption states, classifies results under predetermined acceptance gates, and writes an empirical interface map or scoped null certificate with provenance and uncertainty.
18. The apparatus of claim 17, wherein each network comprises between eight and 256 resonator or oscillator nodes joined by calibrated switched links.
19. The apparatus of claim 17, further comprising independently supported clocks or phase references for the first and second networks and a declared phase-transfer rule across the coupling structure.
20. The apparatus of claim 17, wherein the map writer separately identifies qualification at raw-record, conventional-network-control, cross-registry-custody, physical-retained-state-transfer, and occupied-non-embeddable-interior layers.
21. A relational-interface module comprising an interface coupler, two authenticated address ports, a custody-token processor, phase and timing witnesses, a calibration memory storing a qualified interface map, and a controller configured to create and remove a mapped interface edge only inside a stored operating envelope and to report a validity state and custody reconciliation result.
22. The module of claim 21, wherein a plurality of the modules forms a reconfigurable interior network and a system controller measures an array interface map rather than presuming link independence or additivity.
23. A prophetic occupied-interior apparatus comprising: an exterior threshold housing; an exterior boundary array; a separately maintained interior custody set including plural interior addresses and independent support systems; a phase-addressed interface coupler; a custody controller configured to preserve declared physical invariants of an admitted payload while transferring custody between an exterior address and an interior address; and a support-ledger recorder, wherein an achieved interior operational metric and capacity are outputs of operation and are not fixed limitations of the apparatus.
24. The apparatus of claim 23, wherein the declared invariants include at least identity, whole-turn census, internal clock ratios, bonded adjacency, stored records, orientation where declared, and accounted energy and angular momentum.
25. A prophetic method of operating the apparatus of claim 23, comprising establishing separate exterior and interior address books; phase-locking or establishing a measured phase-transfer rule at the threshold; obtaining co-signed readiness from selected exterior and interior addresses; admitting a nonliving instrument package while continuously recording custody and support ledgers; testing for an ordinary hidden route, substitution, duplication, destructive reconstruction,

and display-only countermodel; releasing the package; and reporting either a custody-preserving transfer, an identified conventional mechanism, or a null.

ABSTRACT

A controlled relational-interior system maintains separate exterior and interior address books and selectively joins them through a mapped interface edge. A controller authenticates both books, binds a unique custody token to an admitted carrier or retained state, commands phase and coupling, monitors bypass channels, capacity, and support accounts, and performs safe release or quarantine on a failed validity condition. Qualification uses randomized sham, bypass, relabeling, duplicate-token, phase-reversal, route-order, delayed-release, and interruption controls to produce an empirical interface map or scoped null certificate. Buildable phase-network, robotic-carrier, topology-control, holonomy, and capacity embodiments are disclosed. A prophetic occupied embodiment preserves custody while transferring a retained organization to a separately maintained interior network whose operational metric or capacity need not be bounded by the exterior displayed volume; no such occupied result is represented as achieved.

Controlled relational-interior system with exterior assembly, interior network, interface coupler, custody registry, controller, witnesses, support ledger, and safe release

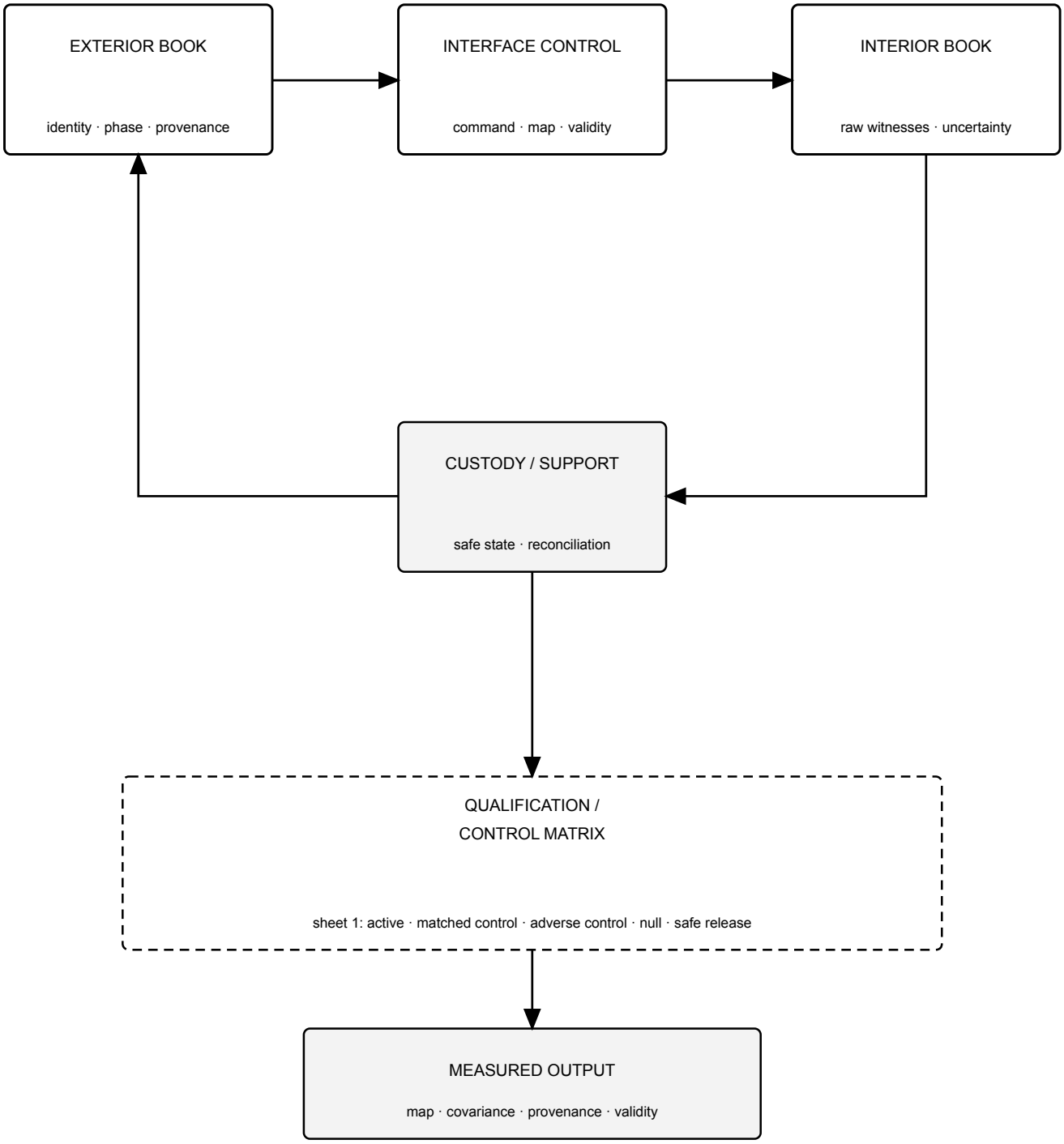


FIG. 1

Controlled relational-interior system with exterior assembly, interior network, interface coupler, custody registry, controller, witnesses, support ledger, and safe release.

Exterior and interior address books joined by a commanded interface edge while remaining separately maintained

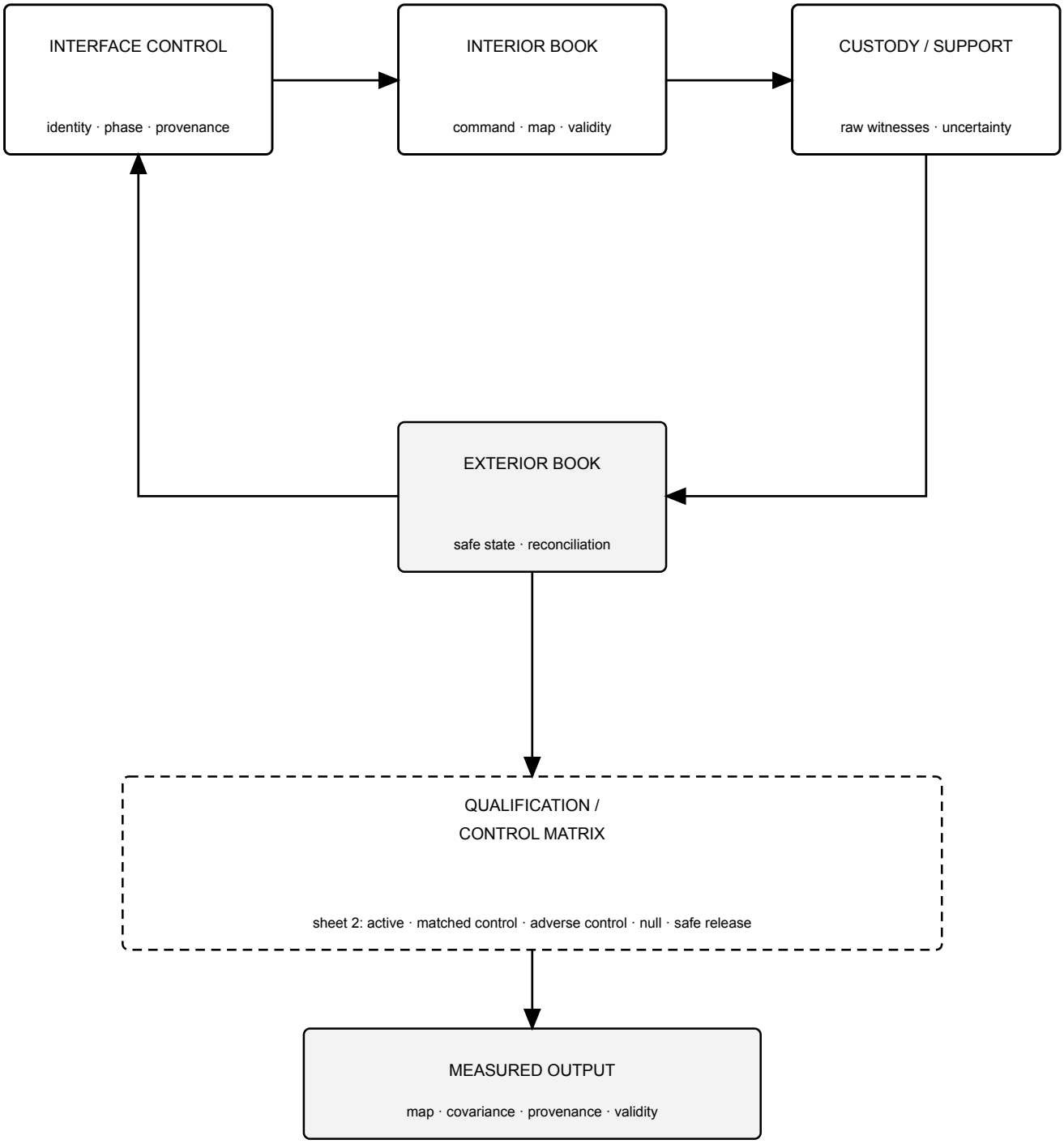


FIG. 2
Exterior and interior address books joined by a commanded interface edge while remaining separately maintained.

Admission, interior-operation, and release state machine with custody continuity and fault quarantine

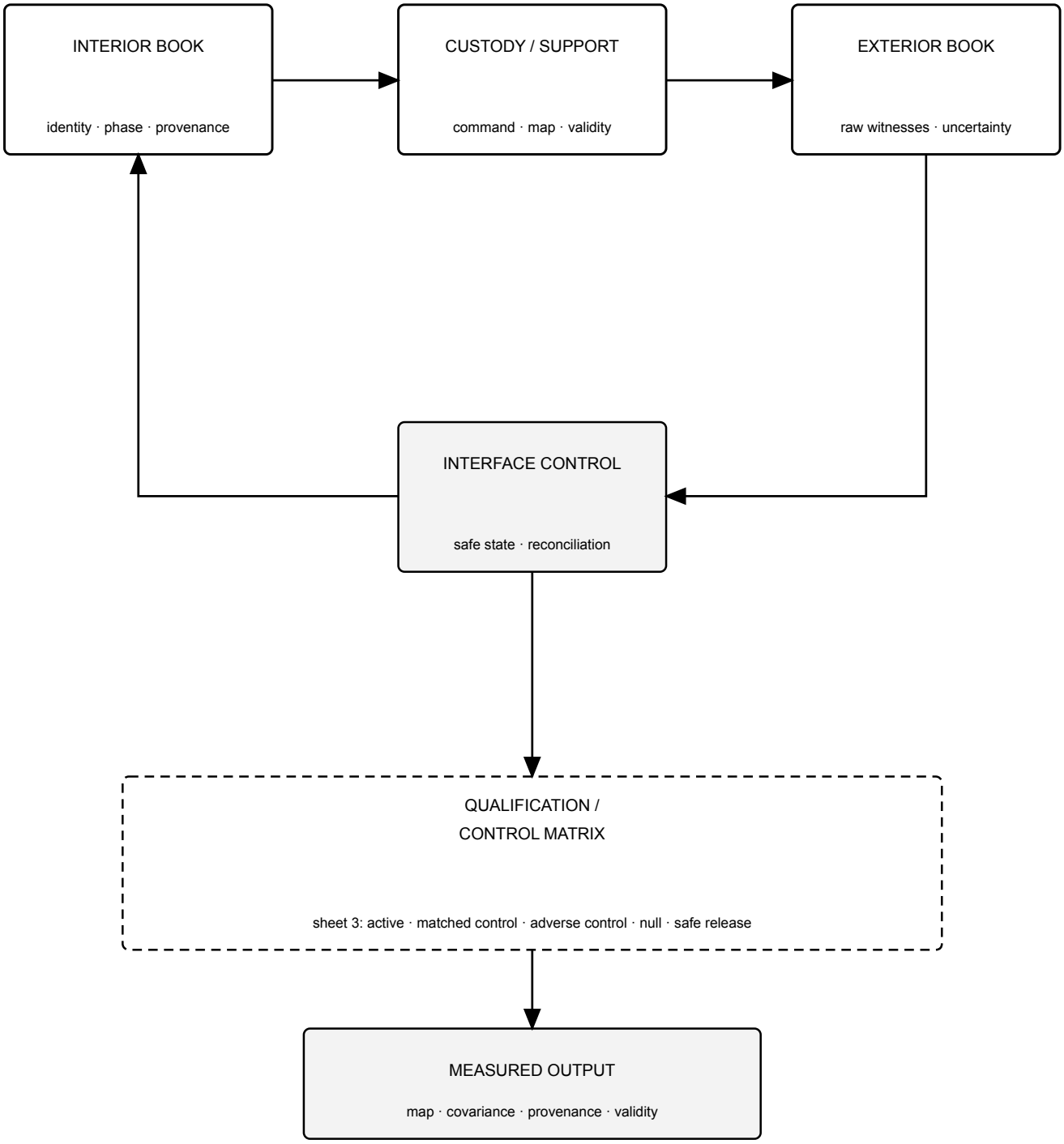


FIG. 3
Admission, interior-operation, and release state machine with custody continuity and fault quarantine.

Weight-control and topology-control modes, route-order comparison, and holonomy receipt

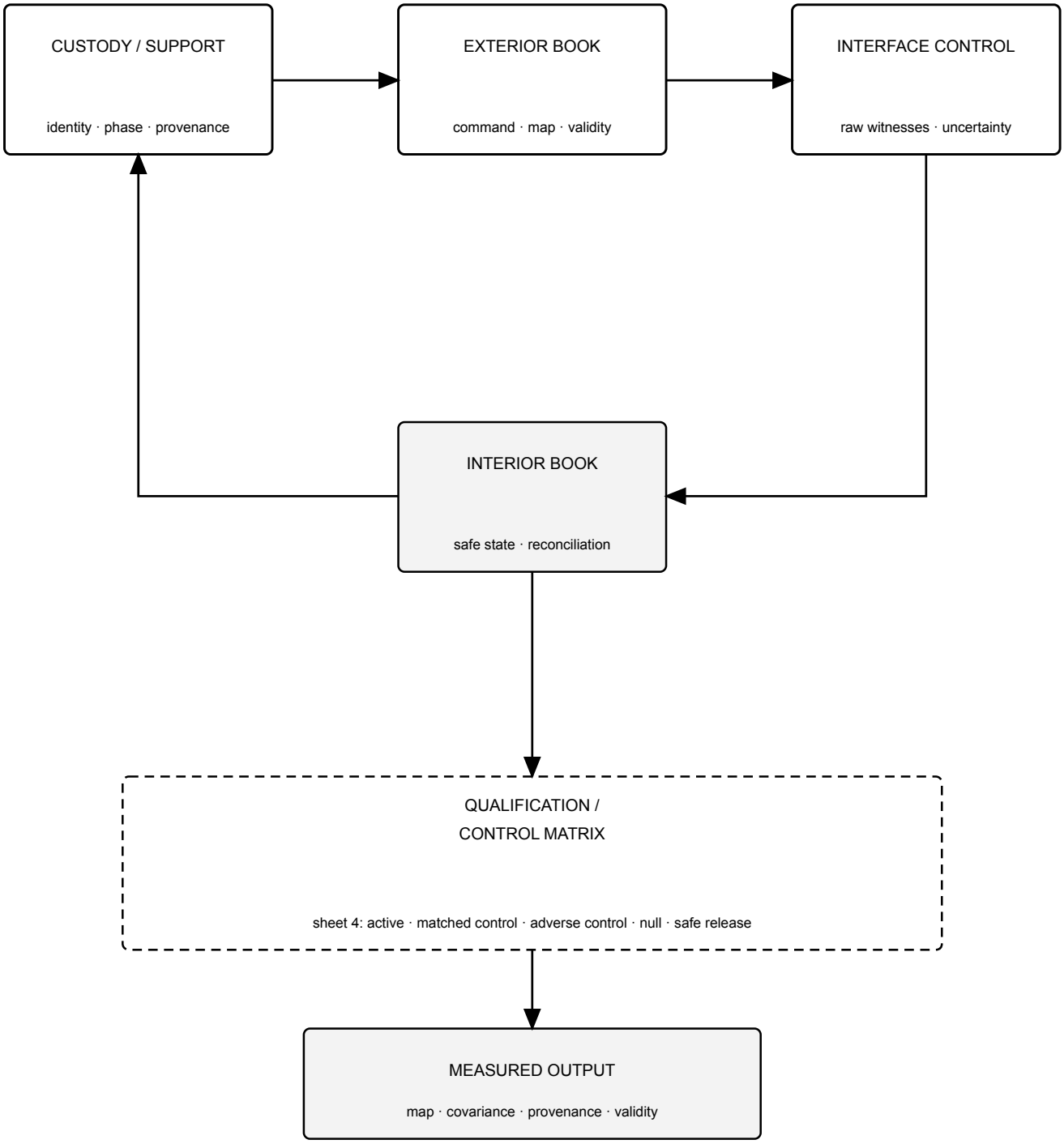


FIG. 4
Weight-control and topology-control modes, route-order comparison, and holonomy receipt.

Finite interface-capacity, occupancy, queue, support-cost, and graceful-degradation account

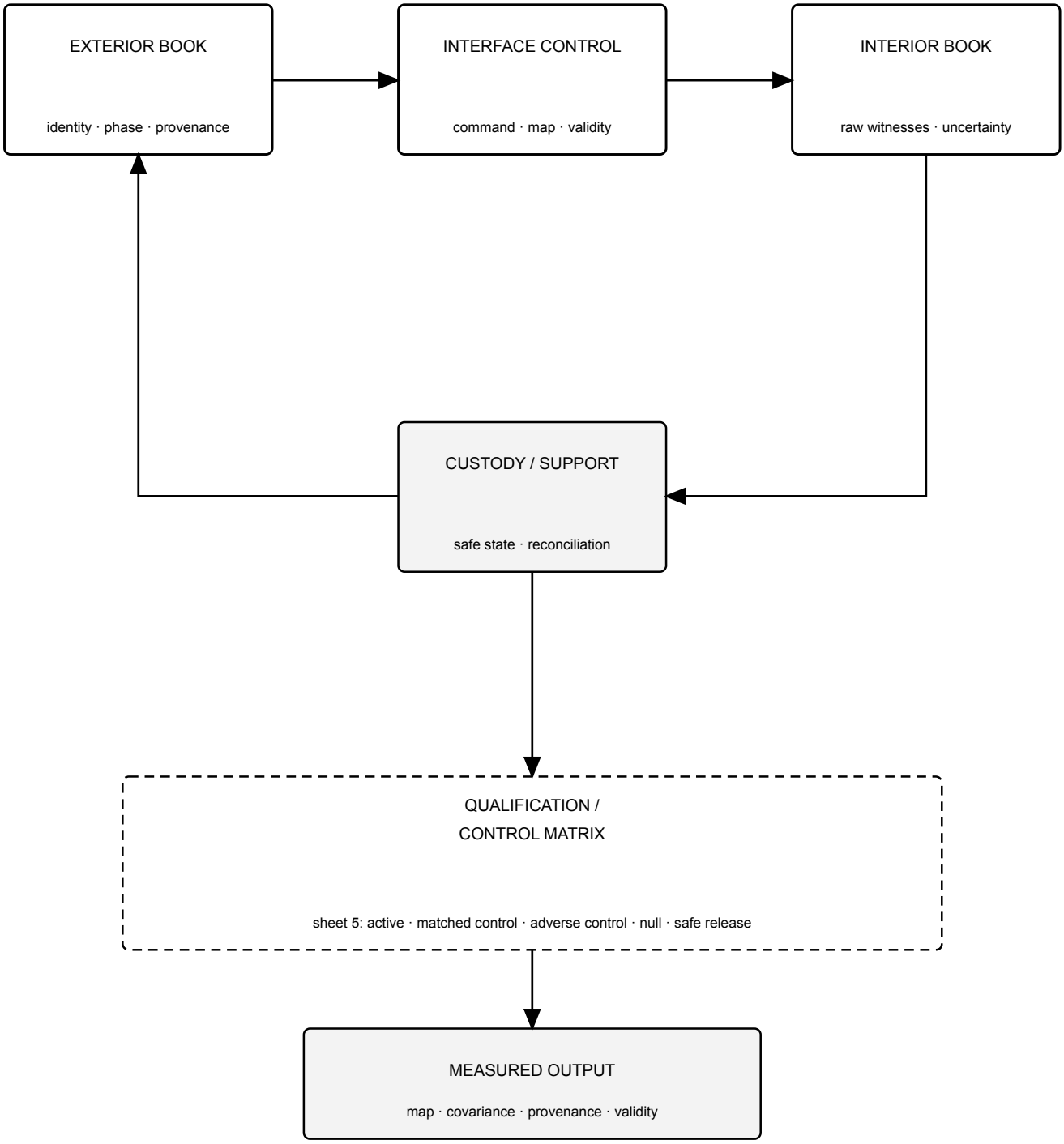


FIG. 5
Finite interface-capacity, occupancy, queue, support-cost, and graceful-degradation account.

Qualification fixture and randomized countermodel matrix

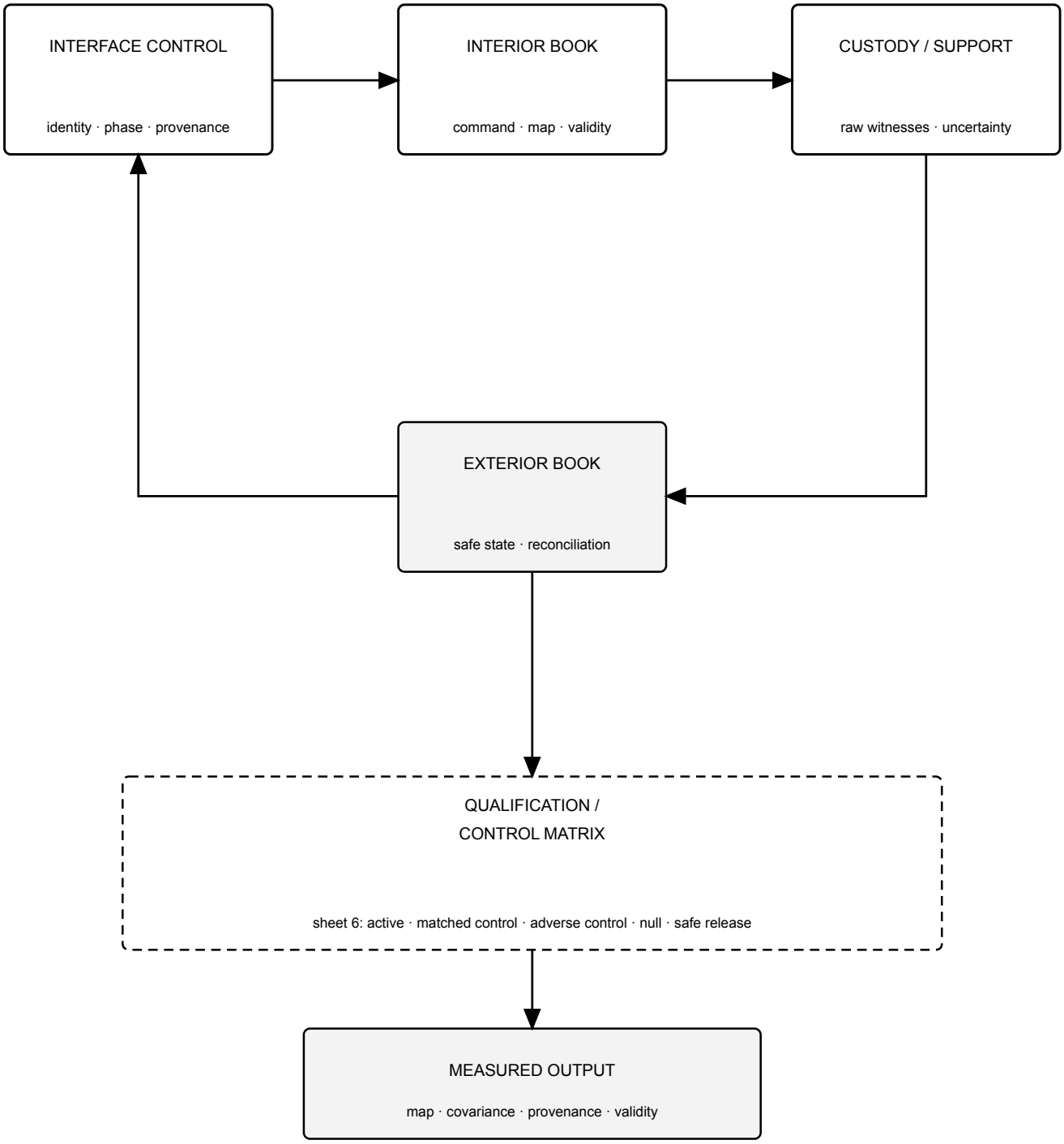


FIG. 6
Qualification fixture and randomized countermodel matrix.

Empirical interface map and layered scope from raw network receipt to occupied prophetic operation

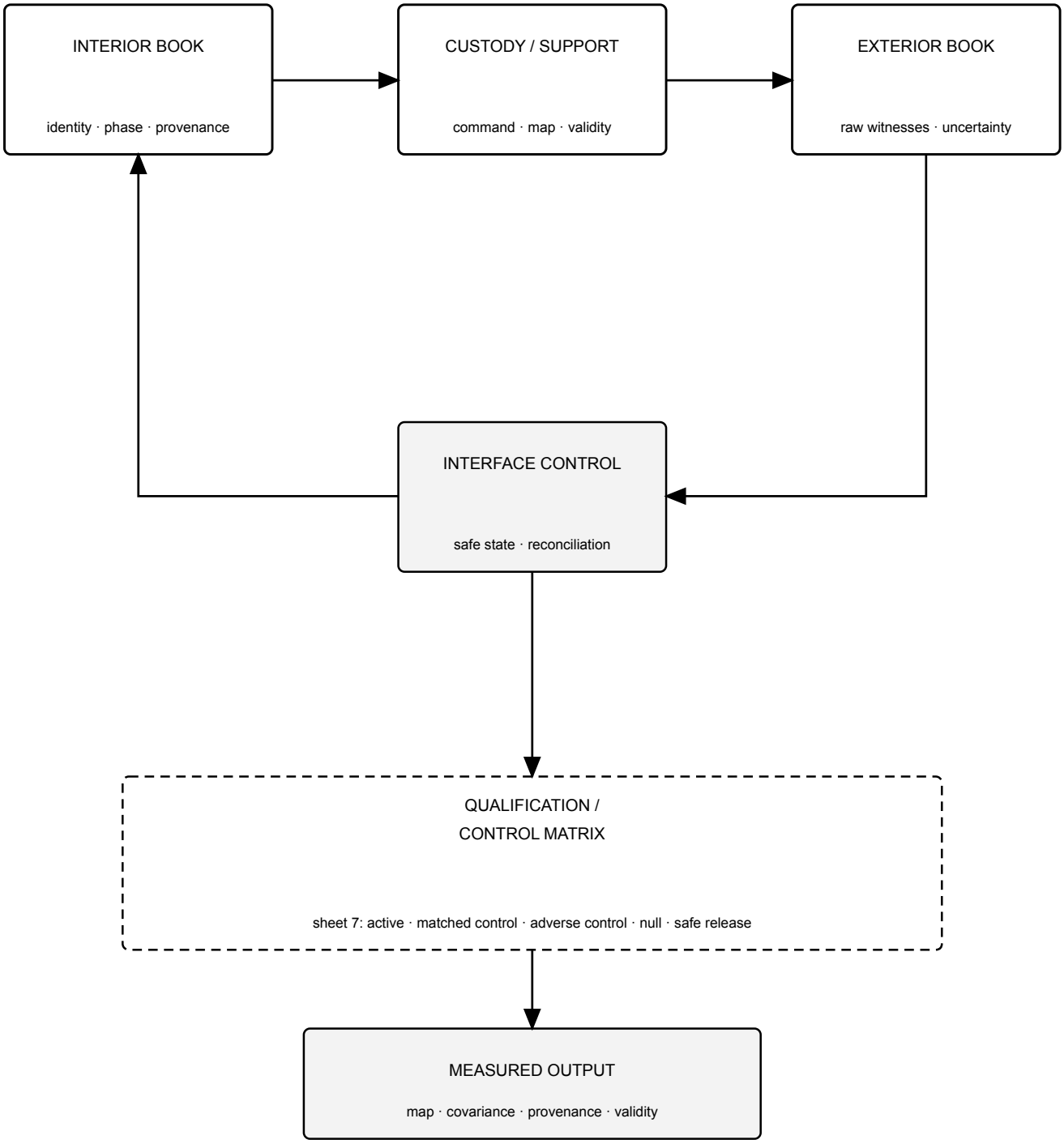


FIG. 7
Empirical interface map and layered scope from raw network receipt to occupied prophetic operation.

Reference phase-network demonstrator using resonator or oscillator nodes, switched links, delay/phase metrology, and tagged carriers

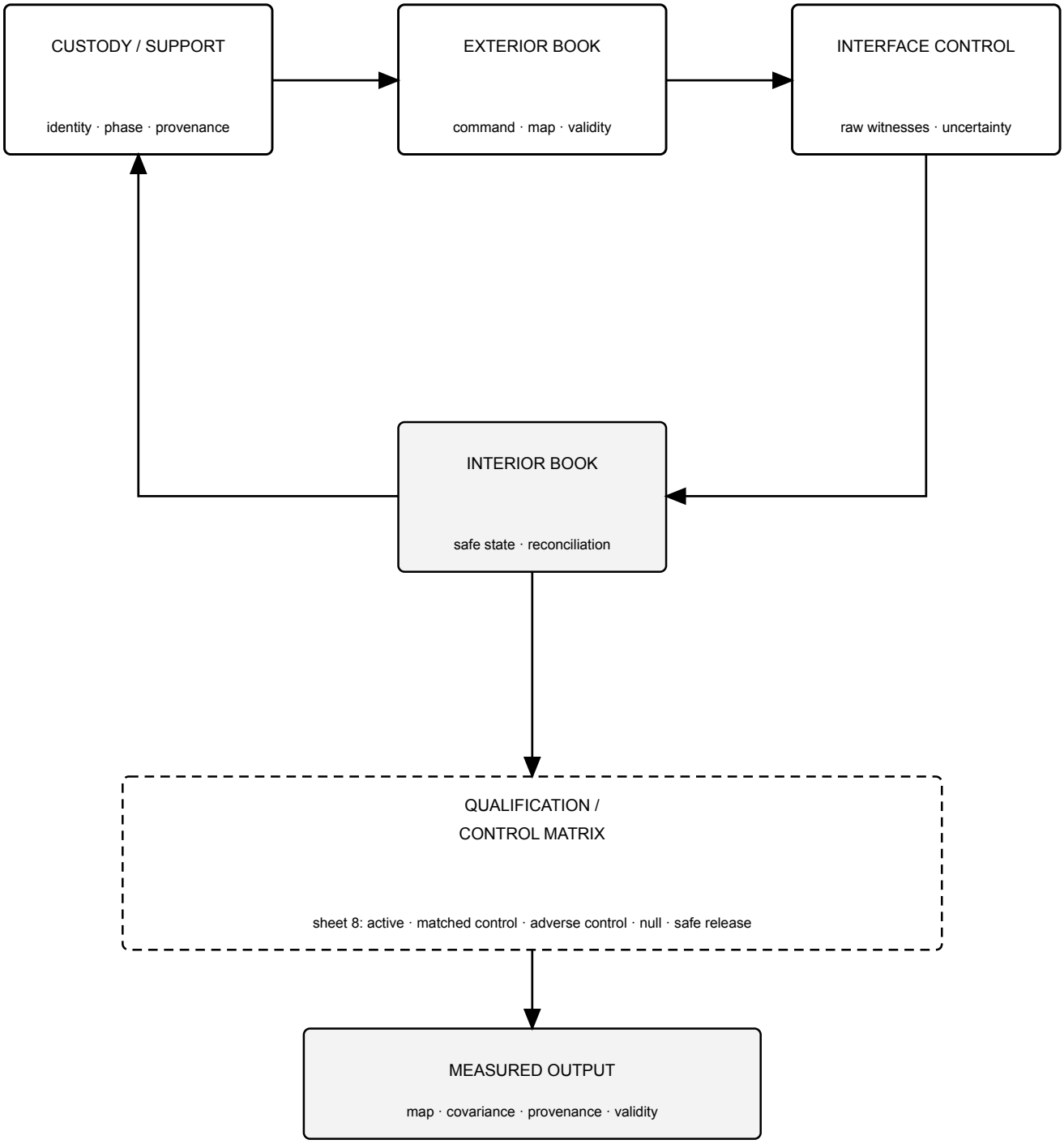


FIG. 8

Reference phase-network demonstrator using resonator or oscillator nodes, switched links, delay/phase metrology, and tagged carriers.

Prophetic occupied threshold, interior custody set, and independent interior support architecture

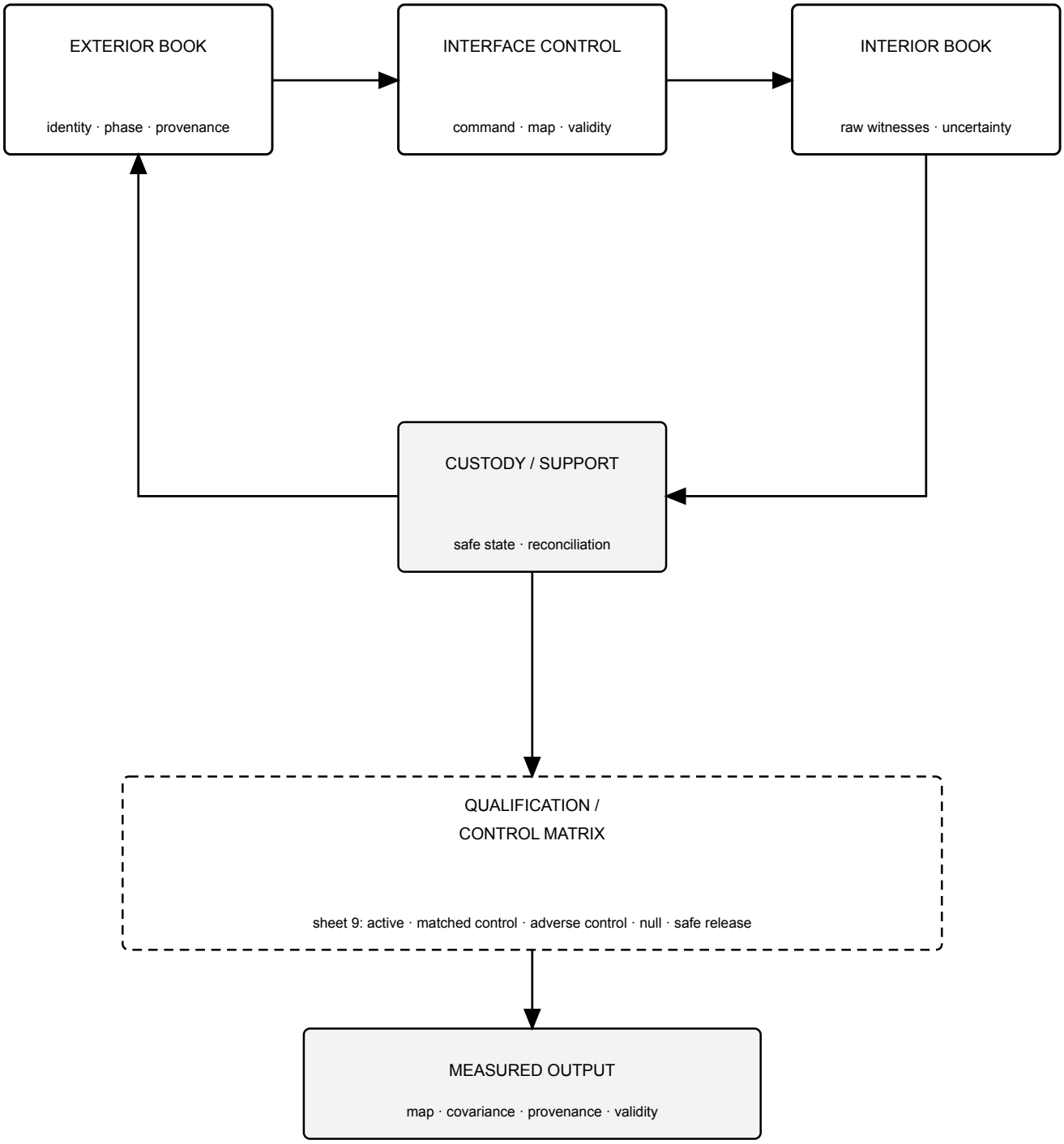


FIG. 9
Prophetic occupied threshold, interior custody set, and independent interior support architecture.

Qualification-to-deployment flow, provenance binding, validity states, and recalibration loop

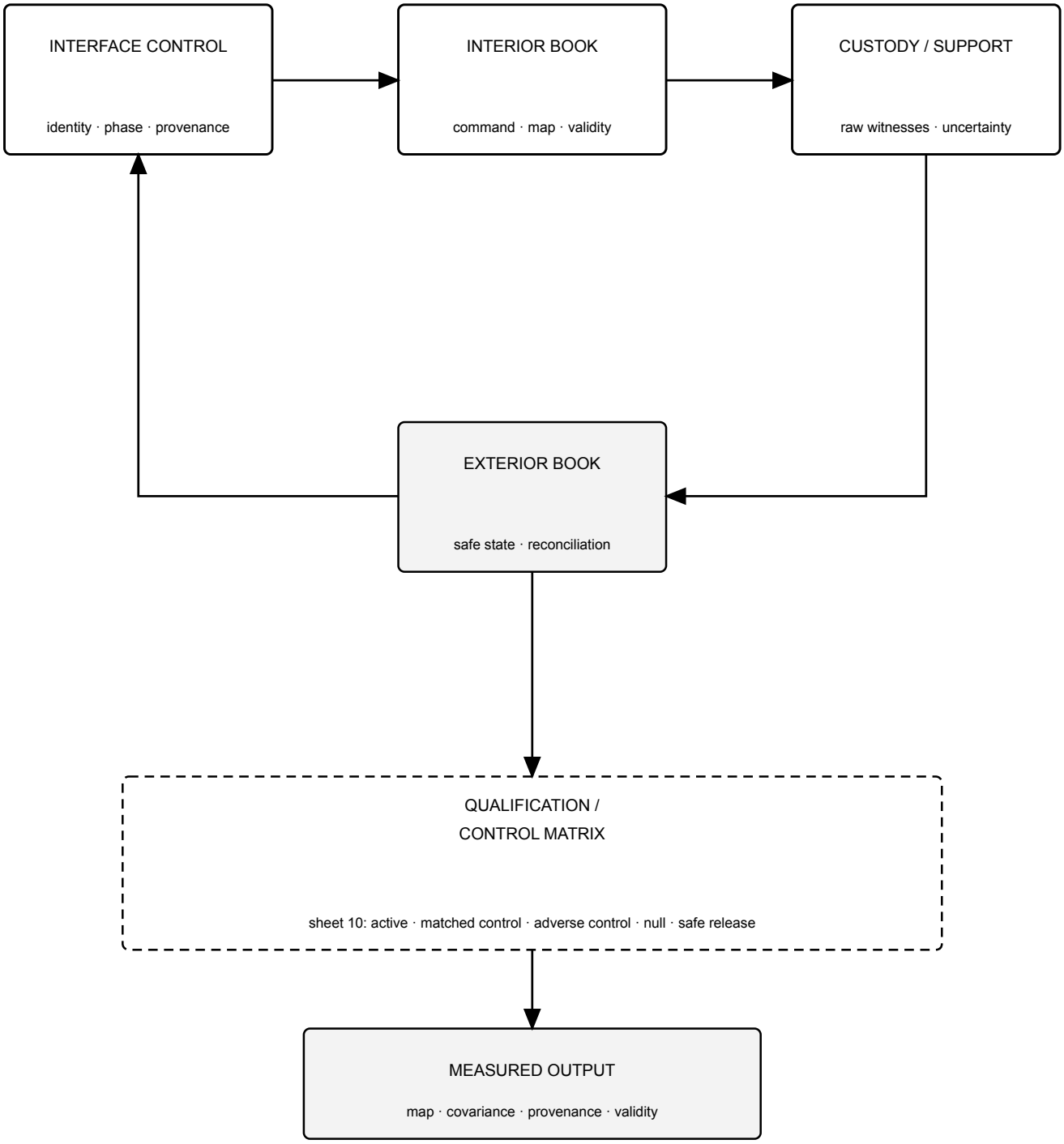


FIG. 10
Qualification-to-deployment flow, provenance binding, validity states, and recalibration loop.